PROJECT REVIEW / EVIDENCE SNAPSHOT

Internship requirements vs. current implementation

AI-Powered Internal Helpdesk Assistant Using RAG

Finalized 25 October 2026, using the latest verified implementation and test evidence

WHAT IS BUILT	EARLIEST OPEN MILESTONE	MAIN REQUIRED WORK LEFT
Text ingestion, chunks, real embeddings, vector search, authentication, corpus administration, exact feedback review, analytics, categories and own profile.	Week 5: connect and exercise a real LLM. Context, prompts, provider adapter, quotation safeguards and tests exist; production generation is disabled.	Employee history UI; formal 30-50-question evaluation; complete retrieval/answer metrics; prompt study; release proof; final documentation, slides and live demo.

How to read this report

BUILT = implemented and verified within stated limits. PARTIAL = code/foundation exists, but behavior or acceptance evidence remains incomplete. PENDING = required work is not delivered. OPTIONAL = not a mandatory blocker. REVIEW = needs a human decision, explanation or approval.

Basis: all 49 sections of the 22-page internship specification, the earlier proposal, current source and retained test artifacts. References use original section/page numbers; E01-E21 identify local evidence. No artificial completion percentage or mentor grade is assigned.

Validation snapshot: 339 backend tests; 34 frontend tests; production build; 14 isolated browser checks and 6 live-app checks. These are software checks, not a measured real-model hallucination rate. No external LLM calls were made for this report.

Reading map

Pages 2-3: weekly progress. Pages 4-10: system requirements. Pages 11-12: mandatory features. Pages 13-17: optional scope, tests, AI metrics, Git and demo. Pages 18-21: delivery, APIs and assessment. Pages 22-24: done gates and next steps. Pages 25-28: evidence index. PDF bookmarks jump to every comparison section.

Week-by-week progress | Weeks 1-6

Specification section 32, pp.13-14; checkpoints section 49, p.22

Work has progressed across several weeks in parallel. The earliest unfinished core milestone is Week 5. A later admin screen does not close the missing live RAG and evaluation gates.

Internship requirement	Current implementation and evidence	Remaining work / interpretation
Week 1: Requirements, AI foundations, architecture and repository	**PARTIAL** Proposal and architecture artifacts exist; stack and modular repository are established. The code demonstrates applied RAG concepts. [E01,E02,E03]	Mentor approval and personal understanding are not verifiable from code. The implementation is now committed; sustained feature-branch/commit discipline still needs improvement.
Week 2: Python/FastAPI, PDF/DOCX/TXT parser, metadata and schema	**BUILT** Working environment, extraction/cleaning, page/heading metadata, SQLAlchemy schema and migrations. Real-store and document fixtures validate the implementation. [E04,E05,E16]	OCR is outside the initial scope. Mentor schema review is a separate sign-off, not evidence of a missing parser.
Week 3: Chunking, embeddings and experiment notes	**BUILT** Tokenizer-aware chunks, overlap, boundaries and embeddings work. Seven fixture configurations have recorded results; UI runs comparisons without code edits. [E06,E07,E17]	Final approved-corpus selection belongs to Week 8. The 200/40 default is provisional, not a universal optimum.
Week 4: Vector store, similarity search and retrieval tests	**BUILT** Chroma, stable PostgreSQL/chunk links, Top-K search, availability checks and retrieval diagnostics work. Generated-corpus results are retained. [E08,E16]	Validate representative approved documents and held-out questions before claiming general retrieval quality.
Week 5: First RAG prototype and initial hallucination tests	**PARTIAL** Context builder, prompt, provider interface/HTTP adapter, quote validation, refusals, citations and injected-provider tests are built. Preview works. [E09,E10]	Select/connect an approved LLM and demonstrate actual supported, paraphrased, misleading, absent-answer and out-of-scope runs. Current provider is disabled.
Week 6: Backend, auth/roles, document/chat/conversation APIs	**PARTIAL** FastAPI, OpenAPI, protected APIs, Argon2id sessions, corpus, chat/history, feedback, review and analytics code are implemented and tested. [E11,E12,E13]	A functional model-backed chat still depends on closing Week 5. Mentor backend/RAG checkpoint approval is not recorded.

Week-by-week progress | Weeks 7-12

Specification section 32, pp.14-16

The admin and analytics work reaches into Week 9, while the employee experience and AI-quality milestones remain partial. This is an engineering status assessment, not elapsed calendar-week accounting.

Internship requirement	Current implementation and evidence	Remaining work / interpretation
Week 7: React/TypeScript web application	**PARTIAL** Login, knowledge base/details, Retrieval lab, Assistant/context preview, experiments, user feedback, Activity, Analytics, Categories and own Profile exist. [E14]	Enable live answers and finish an ordinary employee conversation-history browser/transcript. The Assistant currently shows the current result, not a persistent history screen.
Week 8: 30-50 questions, retrieval/answer evaluation, final settings	**PENDING** Evaluation schemas, evidence labels, comparison UI and fixture reports provide useful groundwork. Four fictional sources have 12 answerable plus 3 unsupported diagnostic questions. [E07,E16,E17]	Create/review the formal 30-50-question corpus-grounded set; measure required retrieval and answer metrics, prompt variants and live-model refusal behavior; select settings on held-out evidence.
Week 9: Feedback analytics, citations, history and admin improvement	**PARTIAL** Aggregate analytics, exact admin turn review, safe citations, UTC filters, categories, profile and feedback audit are built. [E13,E14,E15]	Employee history UI remains. Populate real saved interactions after provider setup; an empty live database correctly produces empty analytics.
Week 10: Functional/API/AI/security/performance tests and release candidate	**PARTIAL** 339 backend tests, 34 frontend tests, build and browser checks passed. Conventional regression and role/upload/store failure paths have substantial coverage. [E18,E19]	Complete live AI behavior, representative end-to-end latency/load checks, a consolidated defect/release report and fresh-install reproduction before declaring a release candidate.
Weeks 11-12: Optional extension and finalization	**PENDING** Local application, migrations, setup guides, source/import handoff, architecture/ERD artifacts and limitations are available. [E02,E03,E20]	Close the core milestones first; package the release, finish technical/user documentation, prepare the 15-20-slide deck and rehearse the complete demonstration. Hosted production readiness is not required.

Purpose, roles and architecture

Specification sections 1-8, pp.2-5

These sections describe the goal and recommended design. Their end-to-end claims remain partial until a real model is connected; recommended libraries are not all mandatory dependencies.

Internship requirement	Current implementation and evidence	Remaining work / interpretation
1. Project overview: enterprise knowledge assistant with sources, feedback, roles and evaluation	**PARTIAL** The surrounding knowledge application, retrieval, metadata, roles, feedback and admin reporting are built. [E04-E15]	Real generated answers and formal retrieval/answer evaluation remain open.
2. Problem statement: find approved internal information using natural language	**PARTIAL** Semantic evidence search addresses paraphrased questions; approved-source availability is enforced. [E08]	Demonstrate employee value and answer quality on a representative approved corpus.
3. Objectives: upload through grounded answer and measurable quality	**PARTIAL** Extraction through retrieval and source provenance work; answer construction and evaluation scaffolding exist. [E04-E10]	Verify actual model-generated outcomes; complete controlled evaluation.
4. High-level ingestion/query/feedback workflow	**PARTIAL** Ingestion, query embeddings, search, context, prompt, persistence and feedback are wired. [E04-E13]	The runtime is retrieval-only with generation_configured=false; no live LLM end-to-end result is claimed.
5. Employee and Helpdesk/Admin roles	**PARTIAL** Both roles log in; employees search/preview and have own-profile access. Admins manage sources, categories, reviews and analytics. [E11,E13-E15]	Employee live answers and history screen remain incomplete. Follow-up submission is not contextual conversational memory.
6. Understand RAG versus a direct LLM chatbot	**REVIEW** The code explicitly retrieves source context before generation; prompts and grounding boundaries are separate services. [E09,E10]	The intern must explain the distinction during mentor review; personal understanding is not proven by generated code.
7. Modular architecture, service boundaries, secrets and backend permissions	**BUILT** Thin routes delegate to services/repositories; providers are isolated; private config is external; role checks are on the backend. [E03,E09,E11]	Maintain these boundaries when integrating the chosen provider. Business logic should remain out of React/route handlers.
8. Recommended technology stack	**PARTIAL** React/TypeScript/custom CSS, Python/FastAPI/Pydantic/Uvicorn, Sentence Transformers, Chroma, PostgreSQL/SQLAlchemy and PyTest are used. [E03,E06,E11,E14]	No live LLM selected. React Router/TanStack Query are not used; navigation/fetch is custom. LangChain, alternative vector stores and Docker are optional, not missing mandatory features.

Document ingestion, chunking and vector storage

Specification sections 9-13, pp.5-7

The document-processing and retrieval foundation is implemented. Validation results are from explicitly fictional and separately identified source fixtures; no OCR capability is implied.

Internship requirement	Current implementation and evidence	Remaining work / interpretation
9. Upload, type/size validation, extraction, metadata and status	**BUILT** PDF with text layer, DOCX and TXT work; Markdown is extra. 20 MiB uploads, request caps, warnings and processing/error status are implemented. [E04,E12]	Scanned or mixed scanned/text PDFs are rejected to avoid silently incomplete extraction. Sophisticated OCR is outside the initial scope.
9. Re-index and deactivate/delete	**BUILT** Admins re-index, replace immutably and toggle online/offline availability. Offline versions are excluded from retrieval but retained for inspection. [E12]	The allowed deactivation path is implemented. Permanent document deletion and a DELETE document route are absent.
10. Pipeline: upload -> clean -> chunks -> embeddings -> indexed	**BUILT** Staging, vector verification and atomic relational publication preserve the previous valid version on failure. Recovery/report tooling exists. [E04,E08]	Processing is synchronous. A durable worker is a scalability enhancement, not a required internship feature.
10.1 Metadata: document/name/type/category/version/page/heading/chunk/date/status	**BUILT** Required metadata is retained where available, plus checksums, source snapshots, offsets and processing fingerprints. PDF physical pages are preserved. [E04,E05]	DOCX/TXT do not gain invented page numbers. Legacy metadata that was not captured remains unknown.
11. Paragraph, sentence and heading boundaries; size/overlap experiments	**BUILT** Token windows prefer meaningful boundaries; page/section-aware overlap and UI-configured size/overlap comparisons exist. [E06,E07,E17]	500/50 is a suggested starting experiment in the PDF, not a required default. MiniLM accepts 256 input tokens, so 200/40 avoids truncation; choose final values by evaluation.
12. Select model; document/query embeddings; retrieval quality	**BUILT** Local all-MiniLM-L6-v2 supplies 384-dimensional document/query embeddings with tokenizer/input-limit checks. Fixture retrieval quality is measured. [E06,E16]	Comparing a second embedding model is optional. Formal approved-corpus quality remains a Week 8 gate.
13. Collection, insertion, metadata, Top-K, re-index/removal	**BUILT** Chroma storage and cosine-based search use stable IDs. PostgreSQL validates active/online source eligibility; cleanup/reconciliation handles retained/stale vectors. [E08]	The two stores do not share a distributed transaction. This limitation is documented and handled through staging/verification/recovery.

Retrieval, prompts, hallucinations and citations

Specification sections 14-17, pp.7-8

Working construction code must be distinguished from evidence of real-model quality. Source quote validation establishes provenance; it is not a proof of semantic correctness or immunity to hallucinations.

Internship requirement	Current implementation and evidence	Remaining work / interpretation
14. Question embedding, vector search, Top-K and metadata filtering	**BUILT** Search and context preview use real embeddings/retrieval and enforce eligible source state. Returned evidence has source location and similarity. [E08,E09]	Validate more paraphrases and difficult policy cases on the approved evaluation corpus.
14. Context assembly -> prompt -> LLM -> answer + sources	**PARTIAL** Bounded context, prompt version/hash, adapter boundary, structured evidence selection, citations and safe persistence are built. [E09,E10]	The provider is disabled. /api/chat returns 503 while unconfigured; no actual LLM-backed employee answer has been validated in this snapshot.
14.1 Top-K experiments, hit rate, missed-source analysis and final config	**PARTIAL** Experiments compare chunk size/overlap/Top-K using source-hit, evidence coverage, MRR and timing. Seven fixture runs are documented. [E07,E17]	Investigate approved-corpus misses and choose final settings with held-out questions. Current exact in-memory comparisons are not a production Chroma performance benchmark.
15. Assistant role, context, question, missing-information rule, concise answer, references	**BUILT** Versioned prompt defines the role, untrusted context boundary, refusal and citation/quote requirements. Preview exposes exact messages for review. [E09,E10]	Prompt variants and readability/relevance of model-selected answers still need live evaluation.
16. Supported, paraphrased, absent, misleading and unrelated-question tests	**PARTIAL** Injected-provider tests cover invalid claims, boundaries, no useful sources, refusal/error handling and source changes. Fixture negatives test retrieval diagnostics. [E10,E18]	Run all five question classes against the chosen real model and approved corpus. Current fixtures cannot establish actual hallucination rate.
16.1 Say insufficient information rather than fabricate	**PARTIAL** Explicit insufficient-evidence paths and quote-validation refusal text are implemented; publication fails safely for malformed provider data. [E10]	Verify that live model/source combinations refuse appropriately and do not return irrelevant but genuine quotations.
17. Source name, page/section and optional chunk reference	**BUILT** Source metadata, exact quotes/offsets, immutable version IDs and authenticated file links are supported in retrieval, answers and admin inspection. [E08-E10,E13]	Real generated responses must still be checked for citation relevance, sufficiency and completeness. No fake DOCX pagination is displayed.

Employee chat and conversation experience

Specification sections 5.1 and 18, pp.3 and 8

Conversation storage and admin review are more complete than the ordinary employee history UI. This distinction is visible in the current source rather than inferred from older handoff summaries.

Internship requirement	Current implementation and evidence	Remaining work / interpretation
New conversation	**PARTIAL** Assistant resets its conversation ID and current result; server creates a conversation when a successful turn is saved. [E09,E14]	Finish a discoverable employee conversation list and persistent transcript. A failed/unconfigured generation does not create a fictitious turn.
Ask a question and display an answer	**PARTIAL** Question form, API, busy/error states and response rendering are built. Context preview works without a model. [E09,E14]	Configure a real provider; current Ask assistant action is disabled while generation is unconfigured.
Display sources	**BUILT** Evidence/source cards show name, version, location, quote and protected source link. Preview includes prompt/context inspection. [E09,E14]	Verify live answer/source alignment; traceable text is not necessarily relevant or complete.
Ask a follow-up question	**PARTIAL** Later submissions can reuse the same conversation ID. The backend keeps explicit question-to-answer links. [E09,E13,E14]	Prior turns are not fed into retrieval/generation. Pronoun-dependent follow-ups require optional memory or query rewriting; do not claim that behavior today.
Helpful/not-helpful feedback	**BUILT** Thumbs up/down, owner-only votes, exact response linkage and creation/update timestamps work; accepted submissions are audited. [E13,E15]	Actual live feedback requires saved model-backed responses. Browser proof used a removed, isolated fictional history fixture.
Recent conversation history	**PARTIAL** Owner-scoped GET conversation list/detail APIs and stored messages exist. Admins have a full user/conversation/turn inspector. [E11,E13]	A dedicated ordinary employee History screen and reloadable transcript are missing. The current Assistant keeps only the displayed result in component state.
Loading and error states	**BUILT** Generation availability, retrieval, context preview, feedback saving and per-screen errors have visible states. [E14,E19]	Continue validating provider timeout/refusal experiences with the selected real model.
18.1 Limited conversation memory	**OPTIONAL** Not implemented. Each question is retrieved independently; conversation ID is a grouping mechanism. [E09]	Optional later extension. If added, test that unrelated history cannot contaminate retrieval or become authoritative source evidence.

Admin, analytics, filtering, feedback and audit

Specification sections 19-24, pp.8-10

The latest work closes the earlier aggregate-analytics, profile, date/version-filter, category-management and feedback-audit gaps. No historical events are invented.

Internship requirement	Current implementation and evidence	Remaining work / interpretation
19. Dedicated corpus administration	**BUILT** Upload/list/details/status/chunks/errors, re-index, replace, deactivate, type/category filters and usage/feedback reporting are present. [E12-E15]	Permanent deletion is not implemented; reversible deactivation satisfies the deactivate/delete option.
20. Total/today/week questions and positive/negative feedback	**BUILT** UTC date-filtered counts, daily activity and current owner ratings are computed from PostgreSQL. Empty history returns zero. [E15]	These are saved-message and current-vote cohorts, not every failed request or counts of vote-submission events.
20. Average response time, FAQs and document usage	**BUILT** Known nonnegative timing average with denominator; case/whitespace-normalized FAQs; distinct response/document-version usage. [E15]	Unknown timings remain NULL. FAQs are not semantic clustering; source usage is not download count.
20. No-useful-retrieval and unsupported/low-confidence questions	**BUILT** Explicit no_eligible_sources and insufficient_context outcomes are reported, with unknown history separate. Saved refusal examples are inspectable. [E15]	This implements the supported/unsupported branch. There is no calibrated confidence score, and missing historical data is not inferred as failure.
21. Name/category/upload date/version/status/question date/rating filters	**BUILT** All listed filters have server support and relevant UI controls. Dates are inclusive UTC; version/date queries expose matching stored document versions. [E12,E13,E15]	Question dates use exact question links. Legacy unlinked answers are excluded by an active question-date filter.
22. Completion/failure/admin/unresolved notifications	**OPTIONAL** Inline statuses/errors exist; no persisted notifications table, notification center or delivery service is built. [E12,E14]	Optional in the specification; do not block the mandatory release on a notification center.
23. Helpful/not helpful, optional comments, timestamp and exact turn	**BUILT** Thumbs-only feedback is the user-approved design. Exact question/response links, current vote and UTC timestamps are retained. [E13,E15]	Written comments are intentionally not collected. Optional comments are not a missing mandatory requirement; votes are not gold labels.
24. Audit actor/action/entity/ID/time/details and examples	**BUILT** Source operations, accepted feedback and category changes write audit events. Feedback/category audit is atomic with the mutation. Activity exposes events to admins. [E12,E15]	Older feedback events are not backfilled. Transcript/prompt text is not copied unnecessarily into audit details.

Database, security, APIs and backend structure

Specification sections 25-29 and 40, pp.10-12 and 18-19

The PDF permits equivalent sessions and describes conceptual data relationships. Different physical table names are documented design choices; independent schema/mentor approval cannot be inferred from passing tests.

Internship requirement	Current implementation and evidence	Remaining work / interpretation
25. Database design and example tables	**PARTIAL** 14 PostgreSQL tables and 114 columns, schema/migrations, UUIDs, FKs, indexes and seed data. Role is a user field; document rows represent immutable versions; messages represent questions/responses. [E05,E20]	These consolidated entities cover core relationships. Persisted notifications are absent/optional. Mentor review before coding is not verified; record the approval and current ERD rationale.
26. User -> conversation -> message; question -> answer -> vote	**BUILT** Ownership, explicit same-conversation question/answer links, vote ownership and stable version/chunk references are enforced. [E05,E08,E13]	Historical answers without valid links remain unknown. Admin review does not guess relationships by message order.
26.1 PostgreSQL/vector-store linkage	**BUILT** Stable document/chunk IDs, source snapshots, versioning and store-verification/reconciliation preserve a traceable link. [E05,E08]	Staging and recovery handle the lack of a cross-store transaction; never claim transactional equivalence between PostgreSQL and Chroma.
27. Login/logout, hashing, sessions and backend RBAC	**BUILT** Argon2id passwords; successful legacy bcrypt login upgrades hashes. Fixed eight-hour revocable opaque cookies use stored digests. Protected routes enforce admin/employee permissions. [E11]	JWT is not used. The main specification permits an equivalent session mechanism; reconcile the earlier proposal wording with the mentor.
28. REST API coverage and chat response contract	**PARTIAL** Auth, documents/detail/upload/reindex, chat/history, feedback, analytics and audit routes exist under /api; answer DTOs include IDs/timing/sources. [E11-E15]	No DELETE /documents/{id}; PATCH availability supplies deactivation. Audit route is /api/audit rather than /audit-logs. Actual model-backed chat remains unconfigured.
29. Routes -> services -> repositories -> stores/external providers	**BUILT** Document, processing, embedding, retrieval, RAG, LLM, conversation, feedback, analytics and audit services are implemented with thin HTTP routes. [E03-E15]	Keep external-provider differences inside adapters; no mandatory LangChain dependency or distributed microservices architecture is required.
40. Secret handling, uploads, permissions, private files and data safety	**BUILT** Environment config, ignored credentials/data, bounded validated uploads, admin-only mutation, protected file access, Argon2id and sanitized audit data. Tests cover ownership/origin/failure behavior. [E11,E12,E18]	Use approved non-confidential documents. Internet deployment still requires HTTPS/secure-cookie/origin configuration; no enterprise identity or production-security certification is claimed.

Required screens and project structure

Specification sections 30-31, pp.12-13

Screens are counted by user-visible behavior. Admin conversation inspection does not substitute for the employee conversation-history requirement.

Internship requirement	Current implementation and evidence	Remaining work / interpretation
30. Login	**BUILT** Dedicated sign-in UI, protected application navigation and session expiration/logout behavior. [E11,E14]	Public registration/password reset/account editing are outside the requested basic scope.
30. Employee Chat	**PARTIAL** Assistant form, context preview, response/citation/vote rendering, provider availability and errors are present. [E14]	Configure a model and verify the real question-to-answer path.
30. Conversation History	**PARTIAL** Owner-scoped backend list/detail plus admin user/turn review are built. [E11,E13]	Missing employee history list and persistent full transcript screen.
30. Admin Document Management	**BUILT** Knowledge base has upload, date/version/name/category/type/status filters, counts and maintenance actions. [E12,E14,E15]	Source deletion remains reversible offline deactivation rather than physical erasure.
30. Document Details/Processing Status	**BUILT** Inspector shows exact chunks, pipeline/errors/warnings, provenance, source/version metadata and stored configuration. [E12,E14]	Synchronous indexing has no live background progress stream.
30. Feedback/Analytics Dashboard	**BUILT** User feedback offers exact turn/evidence inspection; Analytics offers aggregate usage, feedback, timing, FAQs, outcomes and source usage. [E13,E15]	Empty live history is displayed honestly. Live-quality metrics require real saved interactions and separate evaluation.
30. Basic User/Profile	**BUILT** Your profile for both roles shows own identity, session expiry and lifetime activity; safe read-only API. [E15]	No role/identity/password editing is implied by a basic profile.
31. Suggested folder structure	**BUILT** backend/api/services/models/repositories/database/utils, frontend, tests, data, evaluation, README and dependencies are organized accordingly. [E03]	chroma_db is the local vector directory rather than literal vector_db. Gold data directories have no formal approved dataset yet. Suggested names are not an exact-name compliance test.

Mandatory feature checklist | Part 1

Specification section 33, p.16

These are the 17 expressly mandatory feature lines. Built means implemented within its stated limits; it is not a claim that all release or final-demonstration gates are passed.

Internship requirement	Current implementation and evidence	Remaining work / interpretation
LLM integration	**PARTIAL** Provider protocol, explicit HTTP adapter and safe error handling are implemented. [E09,E10]	Configure and exercise an approved actual provider/model; no real call is enabled.
Embedding generation	**BUILT** Real local model embeddings for document chunks and employee questions. [E06]	A second model is optional.
Vector database	**BUILT** Chroma persistence and verified PostgreSQL links. [E08]	No alternative vector database is required.
RAG pipeline	**PARTIAL** Retrieval, bounded context, prompts, provider boundary, grounding and persistence are constructed. [E09,E10]	Demonstrate a complete live-model request, including refusal and citations.
Semantic retrieval	**BUILT** Query embeddings, similarity search, Top-K and online/active eligibility. [E08]	Evaluate broader approved-corpus failure cases.
PDF/DOCX/TXT ingestion	**BUILT** All three required text-bearing formats work; Markdown is an extra. [E04,E16]	Scanned PDF OCR is outside the initial scope.
Document chunking	**BUILT** Exact source slices, meaningful boundaries, tokenizer limits, adjustable size/overlap. [E06]	Final settings need a justified approved-corpus study.
Metadata and source citations	**BUILT** Document/version/page/heading/chunk and quote provenance are retained/displayed. [E05,E09,E13]	Citations still need actual-model relevance/completeness evaluation.
Chat interface	**PARTIAL** Assistant form, answer/source/feedback views and loading/error states exist. [E14]	Enable live answers and finish employee history/transcript navigation.

Mandatory feature checklist | Part 2

Specification section 33, p.16

The main required gaps are actual LLM operation, formal evaluation and final acceptance evidence. Optional features do not compensate for these missing mandatory outcomes.

Internship requirement	Current implementation and evidence	Remaining work / interpretation
Admin document management	**BUILT** Upload, inspect, replace, re-index, deactivate and filter versions. [E12,E15]	Hard delete is not implemented; deactivation is the supported option.
Authentication and role-based authorization	**BUILT** Argon2id, revocable opaque sessions and backend admin/owner enforcement. [E11]	No enterprise identity platform is required.
Feedback	**BUILT** Thumbs up/down, exact response/question links, timestamps and submission audit. [E13,E15]	Optional comments intentionally excluded per user preference.
30-50-question evaluation set	**PENDING** Fixtures and label-validation tooling exist, but not the formal reviewed dataset. [E07,E16,E17]	Create and review 30-50 controlled questions with sources, key facts, evidence and unsupported classes.
Retrieval evaluation	**PARTIAL** Fixture source hit, full-evidence coverage, MRR and query timing are available. [E16,E17]	Add Precision@K/Recall@K and final approved-corpus/held-out results; do not rename hit rate as precision or recall.
Answer-quality evaluation	**PENDING** Grounding test scaffolding and diagnostic storage exist. [E10,E18]	Score real answers for correctness, relevance, groundedness, completeness and unsupported claims; report hallucination rate.
Basic analytics	**BUILT** All required metric categories have explicit saved-data semantics and date filters. [E15]	Real history is currently empty; missing values are not fabricated.
Testing and documentation	**PARTIAL** Substantial software tests, setup/admin guides, test artifacts and handoff are available. [E18-E20]	Finish real AI/performance evaluation, defect/release report, final user/technical package and reproducible handover.

Optional features and intentional exclusions

Specification sections 18.1, 22, 34-35, pp.8-9 and 16-17

These are not mandatory release blockers. Advanced work should follow a stable core RAG flow and measured evaluation, rather than delay them.

Internship requirement	Current implementation and evidence	Remaining work / interpretation
Hybrid search; re-ranking; query rewriting; multi-query retrieval	**OPTIONAL** Not implemented; current path is embedding-based retrieval with source-state filtering. [E08]	Only add after error analysis demonstrates a retrieval problem worth solving.
Context compression; conversation memory	**OPTIONAL** Context is bounded by whole chunks and UTF-8 bytes. No semantic compressor or prior-turn memory exists. [E09]	Byte budgeting is not context compression and is not an exact provider-token guarantee.
Automatic question classification; confidence estimation	**OPTIONAL** Not implemented. Document categories and explicit insufficient-evidence flags are present. [E13,E15]	Document category is not discussion-topic classification; similarity/refusal flags are not calibrated confidence.
Document version comparison; duplicate detection	**OPTIONAL** Versions/provenance are inspectable. Same-source bytes/text/config fingerprints enable safe unchanged-import skips. [E04,E12]	No side-by-side content diff or broad cross-document/semantic duplicate detector.
Document expiry alerts; unanswered-question clustering	**OPTIONAL** Not implemented; admins can inspect unanswered/negative turns and source states. [E13]	No expiry scheduler, topic clusters or alert delivery.
Docker deployment; approved local LLM deployment	**OPTIONAL** Local Uvicorn/React launch is documented. No Docker release or deployed local model is present. [E20]	An approved external API can satisfy LLM integration; local deployment is not mandatory.
Optional notifications and written comments	**OPTIONAL** Inline operation results exist. Feedback intentionally uses thumbs only. [E14,E15]	No notification inbox/delivery; comments are optional and were explicitly deferred by user design.
35. Explicitly excluded initial scope	**SCOPE** No foundation-model training, ChatGPT-scale model, mobile app, enterprise identity, complex approvals, autonomous agents, sophisticated OCR or distributed microservices.	These are intentional exclusions, not missing internship features. Confidential data access still requires approval.

Conventional testing and release evidence

Specification section 36 and Week 10, p.17 and p.15

Recorded tests validate software behavior, isolated stores and browser flows. They do not prove real-model correctness, user adoption or production-scale performance.

Internship requirement	Current implementation and evidence	Remaining work / interpretation
Document-processing and utility unit tests	**BUILT** Extraction order/metadata/negative cases, exact chunks, token limits, boundary/encoding and utility tests. [E04,E06,E18]	Continue regression coverage when parsers or chunk settings change.
API tests; auth/authorization; upload validation	**BUILT** Real PostgreSQL and API tests cover roles/ownership, Argon2id/session behavior, file/request limits and protected routes. [E11,E12,E18]	The suite has one pre-existing Starlette/httpx deprecation warning; no observed failure.
RAG retrieval and configuration-change regression tests	**BUILT** Real local model/store fixtures, source eligibility/version checks and immutable re-indexing/recovery paths. [E08,E16,E18]	A passing fixture set is not the final representative retrieval evaluation.
Prompt/response and unsupported-question tests	**PARTIAL** Deterministic injected-provider tests cover prompt/context, invented/malformed quotes, refusals, errors and source changes. [E10,E18]	Execute and review the chosen real model on supported, paraphrased, absent, misleading and unrelated cases.
Frontend tests and browser behavior	**BUILT** 34 frontend tests and TypeScript/Vite build passed. Latest isolated browser run has 14 checks; six additional live-app checks passed. [E19]	The browser runs made zero chat/provider calls. Isolated fictional transcripts were removed.
Functional/regression backend suite	**BUILT** Latest full run: 339 passed, one existing warning, 198.72 seconds, including opt-in isolated integration tests. [E18]	This is one full-suite total; earlier overlapping totals must not be added together.
Performance checks	**PARTIAL** Query-time measurements and known saved response times exist; average skips missing/negative timing. [E15,E17]	No representative live-model end-to-end latency distribution, load/concurrency study or release budget has been demonstrated.
Defect log, test report and release candidate	**PARTIAL** Regression logs, browser reports, technical notes, backups and handoff record fixes and verification. [E18-E20]	Consolidate a final defect/retest log and acceptance report; verify another-person setup before naming a release candidate.

AI/ML evaluation | Required metric by metric

Specification section 37, pp.17-18; Week 8, p.15

A schema column or user thumbs vote is not an implemented quality metric. Dataset approval, denominators and tested model/configuration must be explicit in the final evaluation report.

Internship requirement	Current implementation and evidence	Remaining work / interpretation
Controlled 30-50 questions with expected sources and key facts	**PENDING** Fixture set has 12 answerable plus 3 unsupported diagnostics; a separate proposal test has 3 questions. Live evaluation tables have no records. [E16,E17,E20]	Freeze an approved/synthetic versioned corpus and independently review 30-50 labels. Do not merge unrelated tests to invent the required dataset.
Retrieval Hit Rate	**PARTIAL** Fixture source-hit and authored evidence-coverage rates are measured. [E16,E17]	Run the reviewed formal set and record answerable/unsupported denominators separately.
Precision@K and Recall@K	**PENDING** Not implemented as formal metrics in the current comparison scorer. [E07]	Define source/chunk relevance sets and compute both metrics. Source-hit and full-quote coverage are not equivalent.
Answer correctness and answer relevance	**PENDING** No scored real-model answer dataset/report. Exact source quote checks are available. [E10]	Compare answers with expected key facts and question intent using a documented human/rubric process.
Groundedness	**PARTIAL** Deterministic exact quote/source/boundary provenance validation is implemented. [E10]	Score semantic support on real answers. A genuine quote can still be irrelevant, incomplete or misleading.
Completeness	**PENDING** Retrieval full-evidence coverage is measured, but generated-answer completeness is not. [E16,E17]	Check all expected key facts/conditions appear in answers; do not substitute retrieval coverage.
Hallucination rate	**PENDING** Unsupported/refusal paths have tests; real-model unsupported-claim frequency has not been measured. [E10,E18]	Define unsupported claims and the denominator; review normal and adversarial/out-of-scope questions.
Average response time	**PARTIAL** Known saved-response timing average and fixture retrieval timing are implemented. [E15,E17]	Measure real complete question-to-answer timings with model, corpus, hardware and sample size documented.
Compare settings and select final configuration	**PARTIAL** Seven size/overlap/Top-K fixture comparisons; source hit, evidence hit, MRR and mean query time. No winner applied. [E17]	Add prompt comparisons, approved held-out questions and live answer quality/cost; document selected settings and failed cases.

Corpus, Git delivery and proposal differences

Specification sections 38-39, p.18; original proposal sections 5-10

The 22-page internship specification is the primary checklist. The earlier proposal is supplementary; implementation differences are disclosed rather than silently marked compliant.

Internship requirement	Current implementation and evidence	Remaining work / interpretation
38. Synthetic or approved non-confidential knowledge base	**BUILT** Fictional travel, VPN/remote access, leave and equipment policies cover PDF/DOCX/TXT/Markdown; sample data and negative fixtures exist. [E16]	Approve/freeze the final representative corpus. The ten suggested sample filenames are examples, not exact required files.
39. Git throughout; meaningful frequent commits	**PARTIAL** Current implementation was committed on 25 October in bfed5eac: Implement helpdesk ingestion, RAG scaffolding and admin workspace. [E21]	The large delivery commit closes the uncommitted-code gap. Frequent focused commits throughout development remain a process improvement.
39. main/develop/feature branch workflow and GitHub	**PARTIAL** Only main is present locally; the feature/develop workflow is not evidenced. Origin was fetched and the current implementation was pushed to origin/main. [E21]	Remote hash verification is recorded with the final push. Keep a reviewed branch/commit workflow for subsequent work; Git sync is distinct from internship feature acceptance.
Proposal: JWT-based sessions	**REVIEW** Current design uses database-backed opaque HttpOnly sessions. Main specification section 27 explicitly allows JWT or equivalent sessions. [E01,E11]	Update the proposal/design decision record or obtain mentor acceptance of the documented session choice; JWT is not a main-spec blocker.
Proposal: Tailwind, React Router and TanStack Query	**REVIEW** React/TypeScript uses custom CSS, local navigation state and a typed fetch client. The main PDF recommends these tools/equivalents. [E01,E14]	Record the simpler implementation choice. Do not claim those particular libraries are installed.
Proposal: separate example tables and no searchable text duplication	**REVIEW** Current schema consolidates roles/versions/Q&A and stores immutable extracted text/chunks in PostgreSQL for evidence and recovery, alongside Chroma. [E01,E05]	Document the rationale in the final architecture/ERD. Confirm mentor approval rather than pretending the original high-level schema is unchanged.
Proposal: DELETE document route	**REVIEW** Current API supplies safe deactivation through PATCH availability; no hard-delete route. [E01,E12]	Document the endpoint variation. Add literal DELETE only if the mentor requires API-contract matching beyond the feature wording.

Final demonstration | What can be shown today

Specification section 41, p.19

The required demonstration must show real end-to-end behavior. Software tests with scripted responses or seeded history are not a substitute for the final live-model demo.

Internship requirement	Current implementation and evidence	Remaining work / interpretation
Admin uploads several knowledge documents	**BUILT** Upload and role validation work for required formats. [E12,E16,E19]	Choose approved demonstration documents and explain their source categories.
Process/index; show status and chunk counts	**BUILT** Knowledge base and inspector show indexing results, exact chunks, provenance and errors. [E12,E14]	Rehearse a clean fresh upload and one expected rejection.
Employee asks a normal question and system retrieves chunks	**BUILT** Retrieval lab and context preview show real matching passages and locations. [E08,E09]	The final normal-question demonstration must also include the real answer path.
LLM produces grounded answer; source/page displayed	**PARTIAL** Prompt, adapter, quote validation and cited answer UI are built. [E09,E10,E14]	Connect a real model and demonstrate correctness/relevance, not only an adapter test.
Employee asks follow-up question	**PARTIAL** A subsequent question can use the current conversation ID. [E09,E14]	Demonstrate the supported independent-question behavior; do not promise contextual pronoun resolution without memory.
Employee votes; admin reviews feedback and analytics	**BUILT** Exact saved-turn votes, evidence inspection, aggregate reporting and feedback audit work in browser fixtures and protected APIs. [E13,E15,E19]	For the final demo, create these records from a genuine completed provider-backed interaction.
Unsupported question does not fabricate an answer	**PARTIAL** Refusal and provenance rejection are implemented and deterministically tested. [E10,E18]	Show the chosen model refusing an unsupported question, and explain limitations of verified quotations.
Intern explains architecture and evaluation results	**PENDING** Technical notes, ERDs and fixture reports support preparation. [E02,E17,E20]	Prepare the final approved-dataset results and rehearse the explanation with the mentor.

Final deliverables | Application and documentation

Specification section 42, p.19

Files existing locally is different from a final reviewed, reproducible submission. This checklist records both what exists and what is needed to close delivery.

Internship requirement	Current implementation and evidence	Remaining work / interpretation
Complete source code	**PARTIAL** Substantial implementation is committed and pushed to the project repository. [E03,E21]	Finish mandatory gaps and prepare a reviewed reproducible release.
Working frontend	**PARTIAL** Built React application and tests; current admin and evidence screens work. [E14,E19]	Finish employee history and real-model chat acceptance.
Working backend	**PARTIAL** Protected FastAPI application, OpenAPI, services/repositories and tests. [E11-E15,E18]	Real chat operation and final acceptance remain incomplete.
RAG pipeline	**PARTIAL** Complete construction through provider boundary and verified quotation publication. [E09,E10]	Connect/exercise a real provider and evaluate outputs.
Vector database setup	**BUILT** Chroma setup, metadata linkage, verification and reconciliation are documented. [E08,E20]	Package the chosen final corpus/index setup without confidential data.
Relational schema and migrations	**BUILT** Current schema, migrations002..006, models and backup runner are present. [E05]	Record mentor approval and include the matching current ERD.
README and installation guide	**PARTIAL** Local environment, migration, account, build/run and test instructions exist. [E20]	Clean stale historical limits; prove a fresh install by another person and document the chosen provider.
User guide	**PARTIAL** Detailed corpus/admin/review/analytics guides exist. [E20]	Finish a unified employee guide for live chat, history, sources, thumbs and errors.
Technical architecture document	**PARTIAL** Proposal, architecture/ERD artifacts, source maps and handoff describe the system. [E01,E02,E20]	Consolidate a current reviewed architecture, store boundaries and deliberate proposal deviations.
API documentation	**BUILT** FastAPI-generated Swagger/OpenAPI plus endpoint behavior in guides. [E11-E15]	Record /api prefix, availability/audit endpoint equivalents and provider-dependent responses.

Final deliverables | Evaluation and handover

Specification section 42, p.19

Use this page as a submission-package checklist after live RAG and evaluation are complete. Supporting source files are indexed at the end of this report.

Internship requirement	Current implementation and evidence	Remaining work / interpretation
Database design	**BUILT** Models/schema/ERD comparison evidence and explicit immutable versions/turn links. [E05,E02]	Explain consolidated entities and obtain/record mentor review.
AI/RAG explanation	**PARTIAL** Pipeline, token limits, context/prompt/quote safeguards and limitations are documented. [E06,E09,E10,E20]	Add actual model choice and evaluated behavior; intern must explain it independently.
Embedding/chunking/retrieval experiments	**PARTIAL** Realistic fixture retrieval tests and seven size/overlap/Top-K comparisons. [E16,E17]	Final approved-corpus study and justified selected configuration are pending; second embedding model is optional.
Prompt experiments	**PENDING** One versioned prompt and deterministic prompt/grounding tests exist. [E10]	Compare prompt variants using actual generated answers and the controlled evaluation set.
Evaluation dataset and results	**PENDING** Fixture data/scoring scaffolding exists, not the formal requirement. [E07,E16,E17]	Deliver the reviewed 30-50 questions, labels, retrieval/answer metrics and error analysis.
Test report	**PARTIAL** Logs and browser reports capture 339 backend/34 frontend tests and recent UI checks. [E18,E19]	Consolidate final software/AI/performance results, known defects and release acceptance.
Known limitations	**BUILT** Extraction, legacy provenance, store boundaries, disabled model and evaluation limits are disclosed. [E20]	Update historical guide statements to match the final implementation before submission.
Future enhancement list	**BUILT** Optional memory/search/quality/deployment improvements and scope exclusions are documented in notes/specification. [E20]	Prioritize them after the core; distinguish future enhancements from mandatory gaps.
Final presentation	**PENDING** No verified final 15-20-slide deck was found in the reviewed project artifacts.	Prepare problem/use case, architecture, pipeline, AI, screens, security, tests/results, challenges, improvements and live demo.

API contract | Required routes and actual equivalents

Specification section 28, p.11

Every route below has the /api prefix in this application. Availability-based deactivation is a supported feature, but is not falsely presented as a literal DELETE endpoint.

Internship requirement	Current implementation and evidence	Remaining work / interpretation
POST /auth/login; POST /auth/logout	**BUILT** /api/auth/login and /api/auth/logout issue/revoke opaque cookie sessions. /api/auth/me exposes safe current identity. [E11]	No JWT token payload; equivalent sessions are permitted by section 27.
POST /documents/upload; GET /documents; GET /documents/{id}	**BUILT** All exist under /api with admin mutation/list permissions and protected document access/detail behavior. [E12]	Date/version filters expose matching historical rows while preserving latest-source behavior without those filters.
POST /documents/{id}/reindex	**BUILT** Explicit re-indexing uses safe immutable version publication. Replacement and availability APIs are also supplied. [E12]	Historical deactivated/staging versions are inspectable, not editable through the UI.
DELETE /documents/{id}	**PARTIAL** No literal DELETE document route. PATCH /api/documents/{id}/availability provides online/offline deactivation. [E12]	Document the variation; physical removal is absent. The feature wording allows deactivation.
POST /chat and response contract	**PARTIAL** /api/chat accepts question, Top-K and optional conversation ID; returns answer/status, response ID, sources and timing when configured. [E09,E11]	Current provider is disabled; the runtime returns 503 rather than a simulated success.
GET /conversations; GET /conversations/{id}	**BUILT** Owner-scoped history list/detail APIs exist under /api. Admin review has separate protected routes. [E11,E13]	Employee history UI is still missing; API support alone does not close that screen requirement.
POST /feedback	**BUILT** /api/feedback validates ownership and assistant response; saves up/down with timestamps and atomic feedback.submitted audit. [E13,E15]	Optional written comments are intentionally excluded.
GET /analytics	**BUILT** /api/analytics is admin-only and accepts strict inclusive UTC dates. [E15]	Saved-data/current-vote reporting, not formal answer-quality scoring.
GET /audit-logs	**BUILT** Equivalent protected route is GET /api/audit, read by the Activity screen. [E12,E15]	Record the route naming difference; no fictitious /api/audit-logs route is claimed.

Presentation, assessment and mentor checkpoints

Specification sections 43-44 and 46-49, pp.19-22

No percentage completion or mentor grade is invented. These requirements include human explanation and approval, which cannot be certified from code or test counts.

Internship requirement	Current implementation and evidence	Remaining work / interpretation
43. Final presentation: approximately 15-20 slides	**PENDING** No verified final deck is present in the reviewed project artifacts.	Cover problem/use case/objectives, architecture/stack, ingestion, embeddings/vector DB/RAG/LLM, screens/security, tests/AI evaluation, challenges/results/future work and live demo.
44. Skills gained	**REVIEW** Artifacts demonstrate practical Python, NLP/embeddings, retrieval, FastAPI/REST, React/TypeScript, PostgreSQL, testing, security and documentation. [E03-E20]	Intern must explain and demonstrate LLMs, RAG/prompts, AI evaluation, Git discipline, debugging and presentation; personal mastery is mentor-assessed.
46. Explain engineering choices and measurable results	**PARTIAL** Chunk limits, overlap/Top-K comparisons, vector storage, prompts, provenance, modularity and limitations have evidence. [E06-E10,E17,E20]	Provide real-model results and confidently explain why RAG, how embeddings/Top-K/prompts work, unsupported behavior and the chosen configuration.
47. Small runnable internal enterprise assistant	**PARTIAL** Reviewer can run a local workspace, upload/manage documents and inspect semantic evidence, protected metadata and reporting. [E12-E15,E20]	Full live-model question/answer and formal evaluation-result paths remain incomplete. Production readiness is explicitly not required.
48. Mentor assessment weights	**REVIEW** AI/ML understanding25%; software engineering20%; application15%; evaluation15%; documentation10%; presentation/demo10%; initiative5%.	These are specification weights, not earned scores. Evaluation and presentation evidence are currently much weaker than software/admin evidence.
49. Mentor checkpoints	**REVIEW** Required reviews: Week1 architecture, Week4 vector/retrieval, Week6 backend/RAG, Week8 AI evaluation, Week10 release candidate, final demo/presentation.	Record actual mentor approval for each checkpoint. Implementation and automated tests do not establish sign-off; Week5/8 gaps affect subsequent acceptance.

Definition of Done | Foundation gates

Specification section 45, pp.20-21

The project is not complete until every required gate is satisfied. Later admin functionality does not waive the missing LLM or controlled evaluation gates.

Internship requirement	Current implementation and evidence	Remaining work / interpretation
Employee can log in	**BUILT** Admin/employee login, cookie session and logout work. [E11,E19]	No core gap for the local scope.
Employee can ask natural-language questions	**PARTIAL** Search/context forms work and the chat request contract exists. [E08,E09,E14]	Live answer submission is disabled until a provider is configured.
Authorized admin can upload documents	**BUILT** Backend role enforcement and bounded type/size-validated uploads. [E12,E18]	No core gap for required supported formats.
PDF/DOCX/TXT content is extracted	**BUILT** Real parser fixtures and isolated validation cover required text-bearing formats. [E04,E16]	OCR remains outside initial scope and is not silently attempted.
Documents are chunked	**BUILT** Exact source slices and model-token-aware boundary handling. [E06]	Final evaluation should justify the selected configuration.
Embeddings are generated	**BUILT** Real cached local model for chunks and queries. [E06,E16]	No second-model comparison is required.
Embeddings are stored in a vector database	**BUILT** Chroma with persisted stable document/chunk identifiers. [E08]	Document setup/recovery as part of the handover.
Relevant chunks are retrieved	**BUILT** Real semantic retrieval and fixture source/evidence checks passed. [E08,E16]	Formal approved-corpus retrieval study is still a separate acceptance task.
LLM generates an answer using retrieved context	**PARTIAL** Construction, prompt, adapter and verified quotation output exist. [E09,E10]	No configured actual model or demonstrated live end-to-end answer.

Definition of Done | Acceptance and delivery gates

Specification section 45, pp.20-21

Use this checklist for the final mentor review. Built items retain their evidence limits; partial and pending gates need explicit closure.

Internship requirement	Current implementation and evidence	Remaining work / interpretation
Sources are displayed	**BUILT** Evidence cards, page/section/version labels, quotes and protected links. [E14]	Check real answer relevance and sufficiency after provider setup.
Unsupported questions handled without fabricated answers	**PARTIAL** No-evidence/invalid-quote refusals and deterministic tests. [E10,E18]	Measure/inspect actual-model behavior; provenance is not semantic correctness.
Feedback can be submitted	**BUILT** Owner thumbs, exact links, UTC dates and audit verified in isolated browser/API runs. [E13,E15,E19]	Normal live usage requires actual saved responses.
Basic analytics are available	**BUILT** New date-filtered Analytics dashboard implements required metric categories. [E15,E19]	Empty history correctly yields zero/unknown values.
30-50 evaluation questions have been tested	**PENDING** Not delivered. Smaller fixture sets are explicitly separate diagnostics. [E16,E17]	Create/review the formal set, run retrieval and real-answer tests, retain results.
AI and software test results documented	**PARTIAL** Software/browser logs and fixture retrieval studies exist. [E16-E19]	Final real-model, full-metric, prompt-comparison and performance report is missing.
Another person can run the README	**PARTIAL** Local run, build, migration, account and test instructions exist. [E20]	Verify a clean setup by another person; include chosen provider and clean committed release.
Final presentation and technical documentation complete	**PENDING** Current guides/architecture/ERD/handoff are useful partial documentation. [E02,E20]	Consolidate current technical/employee guides and prepare/rehearse final deck/demo.

Recommended completion order

Derived action plan; not an additional specification requirement

This order closes the earliest missing milestone, then builds a defensible evaluation and final handover. Provider selection remains deferred until the user is ready; this report makes no API calls.

Internship requirement	Current implementation and evidence	Remaining work / interpretation
1. Finish Week 5: real LLM acceptance	**NEXT** Keep existing context/prompt/provenance architecture; select provider/model and configure secrets locally.	Prove supported/paraphrased/absent/misleading/unrelated questions, citations/refusals and error/timeout handling. Check provider token limits: context currently budgets UTF-8 bytes.
2. Close employee history gap	**NEXT** Reuse owned conversation list/detail APIs and explicit question-answer links.	Add employee history selection, full persistent transcript and reload/resume behavior. Test owner isolation. Contextual memory is a separate optional decision.
3. Freeze corpus and author 30-50 gold questions	**NEXT** Use approved/synthetic versioned documents; reuse source/evidence validators.	Record question ID/type, source identity/version/hash, evidence spans, expected key facts and review status. Keep user thumbs separate and retain held-out questions.
4. Complete formal evaluation and comparison	**NEXT** Reuse one-click size/overlap/Top-K experiments, fixture lessons and prompt versions.	Add Precision@K/Recall@K, correctness/relevance/groundedness/completeness/hallucination measures and full response timings; compare prompts, analyze misses and justify final settings.
5. Harden and package the release	**NEXT** Keep existing role/upload/store regressions and known limitations.	Add realistic end-to-end timing/load checks, consolidated defect/retest report, secure local/deployment configuration, fresh-install proof and release candidate.
6. Document and demonstrate the release	**NEXT** The current application has been committed/pushed; guides still contain historical gaps.	Keep subsequent changes committed, consolidate architecture/user docs, prepare 15-20 slides and rehearse the final demo/mentor checkpoints.

Evidence index | A

Report finalized: 25 October 2026. Latest test evidence: 23 October. Paths are relative unless absolute.

Workspace: C:/Users/HP/Desktop/Sopra_work. This is a code/document review, not a security certification or mentor approval. Git delivery was separately verified after the review. No secrets or private transcripts are reproduced.

Internship requirement	Current implementation and evidence	Remaining work / interpretation
E01 | Primary and supplementary requirements	**SOURCE** C:/Users/HP/Downloads/AI_Powered_Internal_Helpdesk_Assistant_RAG_Internship_Project_Specification_Intern_Version.pdf C:/Users/HP/Downloads/AI_Helpdesk_Assistant_Project_Proposal.docx	49 sections, 22 pages; earlier Week1 proposal. Main specification controls the required checklist.
E02 | Architecture and ERD artifacts	**SOURCE** C:/Users/HP/Downloads/helpdesk_rag_architecture.pdf output/erd_comparison/README.md backend/database/models.py	Earlier diagrams are historical snapshots. Models and latest reflection evidence control the current schema.
E03 | Composition and repository structure	**SOURCE** backend/main.py frontend/package.json requirements_utf8.txt	API composition, dependencies and current stack; service/repository folders are inspected directly.
E04 | Extraction and ingestion	**SOURCE** backend/utils/loaders.py backend/services/document_processing_service.py scripts/validate_real_documents.py	Text extraction, cleanup, immutable snapshots and safe publication.
E05 | Relational model and migrations	**SOURCE** backend/database/models.py backend/database/main_schema.sql backend/database/migrations/006_admin_review_dashboard.sql	All numbered migrations 002..006 inspected through current code/setup evidence.
E06 | Chunking and embeddings	**SOURCE** backend/utils/chunking.py backend/services/embedding_service.py	Tokenizer-aware exact spans, model limits, real local embeddings.

Evidence index | B

Report finalized: 25 October 2026. Latest test evidence: 23 October. Paths are relative unless absolute.

Workspace: C:/Users/HP/Desktop/Sopra_work. This is a code/document review, not a security certification or mentor approval. Git delivery was separately verified after the review. No secrets or private transcripts are reproduced.

Internship requirement	Current implementation and evidence	Remaining work / interpretation
E07 | Experiment construction	**SOURCE** backend/services/experiment_service.py backend/models/experiment_models.py frontend/src/components/Experiments.tsx	Comparison/scoring tools; no live model or automatic final winner selection.
E08 | Semantic retrieval and vector safety	**SOURCE** backend/services/retrieval_service.py backend/services/search_service.py scripts/verify_stores.py scripts/reconcile_stores.py	Chroma query path plus PostgreSQL source-state/provenance eligibility and recovery.
E09 | RAG/context/provider construction	**SOURCE** backend/services/rag_service.py backend/services/context_service.py backend/services/llm_service.py	Disabled production provider, explicit HTTP adapter and UTF-8-byte context budget.
E10 | Prompt and grounding boundary	**SOURCE** backend/prompts/rag_prompts.py backend/services/grounding_service.py tests/test_rag_service.py tests/test_llm_service.py	Exact complete source quotations and deterministic tests; not actual-model quality evidence.
E11 | Auth and conversation APIs	**SOURCE** backend/utils/security.py backend/services/auth_service.py backend/api/dependencies.py backend/api/rag.py	Argon2id, opaque sessions, backend permissions and owner-scoped history.

Evidence index | C

Report finalized: 25 October 2026. Latest test evidence: 23 October. Paths are relative unless absolute.

Workspace: C:/Users/HP/Desktop/Sopra_work. This is a code/document review, not a security certification or mentor approval. Git delivery was separately verified after the review. No secrets or private transcripts are reproduced.

Internship requirement	Current implementation and evidence	Remaining work / interpretation
E12 | Corpus/admin/audit operations	**SOURCE** backend/api/documents.py backend/services/document_service.py backend/api/audit.py backend/services/audit_service.py	Upload, replacement, re-index, availability, protected details and Activity.
E13 | Admin exact-turn/feedback review	**SOURCE** backend/services/admin_review_service.py backend/repositories/admin_review_repository.py backend/services/feedback_service.py frontend/src/components/UserFeedback.tsx	Valid exact Q/A links, evidence, current votes, timestamps and owner review.
E14 | Current React screens/navigation	**SOURCE** frontend/src/App.tsx frontend/src/components/Assistant.tsx frontend/src/components/KnowledgeBase.tsx frontend/src/components/DocumentDrawer.tsx	No employee History screen; Assistant retains its current answer/conversation ID in state.
E15 | Latest analytics/profile/category/filter/audit work	**SOURCE** ADMIN_ANALYTICS_PROFILE_GUIDE.md backend/services/analytics_service.py backend/services/category_service.py backend/services/profile_service.py backend/utils/date_ranges.py	Matching API/repository/React layers and tests were inspected. Current state supersedes older missing-feature notes.
E16 | Realistic document retrieval validation	**SOURCE** output/real_document_validation/VALIDATION_REPORT.md output/actual_proposal_validation/results.json	Four fictional policies: 30 chunks, 12/12 correct source at rank 1, full evidence 9/12 at rank 1 and 12/12 within top 3. Separate proposal 3-question check; not pooled.

Evidence index | D

Report finalized: 25 October 2026. Latest test evidence: 23 October. Paths are relative unless absolute.

Workspace: C:/Users/HP/Desktop/Sopra_work. This is a code/document review, not a security certification or mentor approval. Git delivery was separately verified after the review. No secrets or private transcripts are reproduced.

Internship requirement	Current implementation and evidence	Remaining work / interpretation
E17 | Fixture comparative study	**SOURCE** output/week5_validation/COMPARATIVE_STUDY.md output/week5_validation/comparative_study.json	Seven settings on 12 answerable+3 unsupported diagnostics. Exact in-memory retrieval, no LLM, no final configuration applied.
E18 | Latest backend regression record	**SOURCE** output/admin_completion_validation/final_backend_tests.txt tests/test_analytics.py tests/test_category_profile_filters.py tests/test_feedback_audit.py	339 passed, one existing Starlette warning; includes isolated real-store coverage.
E19 | Frontend/build and browser record	**SOURCE** output/admin_completion_validation/browser_report.json output/admin_completion_validation/live_smoke_report.json output/admin_completion_validation/fixture_cleanup.json frontend/tests/adminData.test.mjs	14 isolated+6 main-app checks; zero page/chat errors/calls. 34 frontend tests/build recorded in latest guide/handoff.
E20 | Current setup, handoff and database snapshot	**SOURCE** README.md ADMIN_REVIEW_GUIDE.md ADMIN_ANALYTICS_PROFILE_GUIDE.md LLM_HANDOFF.txt output/admin_completion_validation/schema_after.json	14 tables/114 columns; 2 users,10 document versions, 17 chunks, no saved chat/feedback/evaluation rows at recorded snapshot. Earlier guide limits can be stale.
E21 | Git delivery snapshot, 25 October 2026	**SOURCE** Local Git commands: branch --list; log -5; status --short	Implementation commit bfed5eac; main only locally. Origin was fetched and the current code pushed. Secrets, uploads, backups, builds and temporary files were excluded; document binaries are protected from newline conversion.